

SUPPLEMENTARY MATERIAL FOR FOURIER ANALYSIS

Pritam Chandra

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1 Some rules of interchanging limits

Definition 1. A function $f: A \rightarrow \mathbb{R}$ is said to be in $C^k(A)$ if f is k times differentiable on A and its k th derivative $f^{(k)}$ is continuous.

Fact 1. *A differentiable function is continuous, but need not be uniformly continuous. An example of such a function is $f(x) = x^2$.*

Fact 2. *Every continuous function on a compact interval is uniformly continuous.*

Rule 1. (Leibniz's rule for differentiating under the integral sign) *Let $f \in C^1(\mathbb{R}^2)$. Then*

$$\frac{d}{dt} \int_a^b f(x, t) dx = \int_a^b \frac{\partial}{\partial t} f(x, t) dx.$$

Rule 2. (Fubini's theorem for interchanging the order of integration) *Let R be the rectangle $[a, b] \times [c, d]$. Let f be a function defined on R . Consider the integrals*

$$\begin{aligned} & \int_R f(x, y) dA(x, y), \\ & \int_c^d \int_a^b f(x, y) dx dy, \\ & \int_a^b \int_c^d f(x, y) dy dx. \end{aligned}$$

If any of the following conditions,

- (i) f is continuous everywhere on R
- (ii) f is continuous everywhere except on a "small" set and f is bounded
- (iii) $f \geq 0$ and one of the three integrals is finite
- (iv) $\int_R |f| dA$ is finite,

then all the three integrals equal.

Rule 3. Let $f: [a, b] \rightarrow \mathbb{R}$ be continuous. Then for all $x \in (a, b)$,

$$\frac{d}{dx} \int_a^x f(t) dt = f(x).$$

Rule 4. (Interchanging sums and integrals) Let $f_n: \mathbb{R} \rightarrow \mathbb{R}$ be a sequence of functions. Then the equality

$$\int_{\mathbb{R}} \sum_{n=1}^{\infty} f_n(x) dx = \sum_{n=1}^{\infty} \int_{\mathbb{R}} f_n(x) dx$$

holds if one of the following conditions hold:

- (i) $f_n(x) \geq 0$ for all $x \in \mathbb{R}$
- (ii) the series $\sum \int_{\mathbb{R}} |f_n(x)| < \infty$.

Rule 5. [Lebesgue dominated convergence theorem (LDCT)] Let f_n and f be a sequence of measurable functions on the measure space $(X, \mathcal{B}, \mu)$ such that $f_n \rightarrow f$ pointwise. Further, suppose f_n is dominated by an integrable function g , that is,

$$|f_n(x)| \leq g(x) \text{ for all } x \in X \text{ and } n \in \mathbb{N}.$$

Then f_n and f are integrable, and

$$\lim_{n \rightarrow \infty} \int_X f_n d\mu = \int_X f d\mu.$$

Rule 6. (Weierstrass M-test) Let f_n be a sequence of real or complex valued functions defined on A . Suppose for each $n \in \mathbb{N}$ there exists a number M_n such that

$$|f_n(x)| \leq M_n \text{ for all } x \in A,$$

and the series $\sum M_n$ is convergent. Then the series $\sum f_n(x)$ converges absolutely and uniformly on A .

2 Tauber's theorem

Theorem 1. [Tauber] Suppose the series $\sum x_n$ is Abel summable and the sequence $n x_n$ converges to 0. Then the series $\sum x_n$ is convergent.

Proof. Let L be the Abel sum of $\sum x_n$. For $0 < r < 1$, let $L(r) = \sum_{n=1}^{\infty} x_n r^n$. For such an r , note that

$$(1 - r^n) = (1 - r)(1 + r + r^2 + \cdots + r^{n-1}) \leq n(1 - r).$$

For any $m \in \mathbb{N}$, let s_m be the partial sums of the series $\sum x_n$ and let $r_m = 1 - 1/m$. We have

$$|s_m - L| \leq |s_m - L(r_m)| + |L - L(r_m)|$$

and

$$|s_m - L(r_m)| = \left| \sum_{n=1}^m x_n - \sum_{n=1}^{\infty} x_n r_m^n \right| \leq \left| \sum_{n=1}^m x_n (1 - r_m^n) \right| + \left| \sum_{n=m+1}^{\infty} x_n r_m^n \right|.$$

Pick any $\varepsilon > 0$. Firstly, since nx_n converges, it is bounded. That is, there is some B such that $|nx_n| \leq B$ for all n . There also exists N_1 such that $|nx_n| < \frac{\varepsilon}{6}$ for every $n \geq N_1$. Further, there exists $N_2 \geq N_1$ such that $\frac{1}{N_2} < \frac{\varepsilon}{6BN_1}$. So, for all $m \geq N_2$

$$\begin{aligned} \left| \sum_{n=1}^m x_n (1 - r_m^n) \right| &\leq \sum_{n=1}^m |nx_n (1 - r_m)| \\ &= \frac{1}{m} \sum_{n=1}^{N_1} |nx_n| + \frac{1}{m} \sum_{n=N_1+1}^m |nx_n| \\ &< \frac{N_1 B}{m} + \frac{\varepsilon}{6} \\ &\leq \frac{N_1 B}{N_2} + \frac{\varepsilon}{6} \\ &< \frac{\varepsilon}{3}, \end{aligned}$$

and

$$\left| \sum_{n=m+1}^{\infty} x_n r_m^n \right| \leq \sum_{n=m+1}^{\infty} |x_n| r_m^n < \frac{\varepsilon}{6} \sum_{n=m+1}^{\infty} \frac{r_m^n}{n} < \frac{\varepsilon}{6m} \frac{r_m^{m+1}}{(1 - r_m)} < \frac{\varepsilon}{3}.$$

Here for the second inequality, the bound $|nx_n| < \frac{\varepsilon}{6}$ has been used again noting that $n > m \geq N_1$. Finally, since $L(r_m)$ converges to L , there exists $N_3 \geq N_2$ such that for all $m \geq N_3$

$$|L(r_m) - L| < \frac{\varepsilon}{3}.$$

So, for all $m > N_3$ we have

$$\begin{aligned} |s_m - L| &\leq \left| \sum_{n=1}^m x_n (1 - r_m^n) \right| + \left| \sum_{n=m+1}^{\infty} x_n r_m^n \right| + |L - L(r_m)| \\ &< \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} \\ &= \varepsilon. \end{aligned}$$

Therefore, L is the sum of series $\sum x_n$. □

3 Cesàro convergence of the Fourier series

The operation $*$, called *convolution*, is a binary operation on $C[-\pi, \pi]$. Suppose $f, g \in C[-\pi, \pi]$. Identify f and g with their extensions to $\mathbb{R}$ with the understanding that $f(\theta) = f(\theta + 2\pi)$ and $g(\theta) = g(\theta + 2\pi)$. Then

$$(f * g)(\theta) = \int_{-\pi}^{\pi} f(t) g(\theta - t) dt.$$

The operation $*$ is commutative. There is no identity element corresponding to this operation. Although, there exist families Q_n such that

$$\lim_{n \rightarrow \infty} Q_n * f = f.$$

Such a family is called an *approximate identity* for convolution. A special approximate identity is the Dirac family.¹

For $f \in C[-\pi, \pi]$, define

$$\hat{f}(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(t) e^{-int} dt.$$

This is always defined as

$$\left| \frac{1}{2\pi} \int_{-\pi}^{\pi} f(t) e^{-int} dt \right| \leq M,$$

where M is the maximum value taken by the continuous function $|f(t)|$ in the compact interval $[-\pi, \pi]$. Define the series

$$\mathcal{F}f(t) = \sum_{n=-\infty}^{\infty} \hat{f}(n) e^{int}.$$

Thus, for every $f \in C[-\pi, \pi]$, the series $\mathcal{F}f$ is defined. This is called the *Fourier series* of f . The numbers $\hat{f}(n)$ are called *Fourier coefficients*.

Question 1. When is it true that $\mathcal{F}f(t) = f(t)$?

On $[-\pi, \pi]$ define the family of functions

$$D_n(t) = \frac{1}{2\pi} \sum_{k=-n}^n e^{ikt} \text{ and } F_n(t) = \frac{1}{n} \sum_{k=0}^{n-1} D_k(t).$$

These families are called the *Dirichlet kernel* and the *Fejér kernel* respectively. The Fejér kernel is a Dirac family, whereas the Dirichlet kernel is not (it can take negative values).

1. A Dirac family is a sequence $Q_n \in C[-\pi, \pi]$ that satisfy four conditions: the functions are (i) positive, (ii) even, (iii) normalized (their integral is 1), and (iv) “peaking at 0”, that is, for any δ -neighborhood of 0, for large enough n , the integral of Q_n outside the neighborhood is arbitrarily small.

Now, for any series $\mathcal{S} = \sum x_n$, let $s_n \mathcal{S}$ be the sequence of partial sums and let $\sigma_n \mathcal{S}$ be the sequence of Cesàro sums. That is,

$$s_n \mathcal{S} = \sum_{k=-n}^n x_k \text{ and } \sigma_n \mathcal{S} = \frac{1}{n} \sum_{k=0}^{n-1} s_k \mathcal{S}.$$

It turns out that

$$s_n \mathcal{F}f = D_n * f \text{ and } \sigma_n \mathcal{F}f = F_n * f.$$

Since F_n is a Dirac family, we immediately get Fejér's theorem.

Theorem 2. [Fejér] *For every $f \in C[-\pi, \pi]$, its Fourier series is Cesàro summable to f uniformly. (Alternatively, f is a uniform limit of trigonometric polynomials.)*

Theorem 3. [Weierstrass approximation theorem] *Let $g \in C[-1, 1]$. Then for every $\varepsilon > 0$, there exists a polynomial p such that*

$$\sup_{x \in [-1, 1]} |g(x) - p(x)| < \varepsilon.$$

That is, polynomials are dense in the Banach space $C[-1, 1]$ with the supremum norm.

Lemma 1. *For every $n \in \mathbb{N}$, there exists a polynomial $T_n(x)$ such that*

$$\cos nx = T_n(\cos x).$$

Proof. We first note that $T_1(t) = t$ and $T_2(t) = 2t^2 - 1$. Note that

$$\begin{aligned} & \cos(n+1)x + \cos(n-1)x \\ = & \cos nx \cos x - \sin nx \sin x + \cos nx \cos x + \sin nx \sin x. \end{aligned}$$

Inductively, let

$$T_{n+1}(t) = 2tT_n(t) - T_{n-1}(t).$$

which is a polynomial satisfying $T_{n+1}(\cos x) = \cos(n+1)x$. □

Below we give a proof of Weierstrass approximation theorem using Fejér's theorem.

Proof. Let $g \in C[-1, 1]$. Consider $f: [-\pi, \pi] \rightarrow \mathbb{R}$ such that

$$f(\theta) = g(\cos \theta).$$

Then $f \in C[-\pi, \pi]$. Further, f is even. Recall that

$$\hat{f}(-n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(t) e^{int} dt.$$

Substitute $u = -t$ and use $f(-u) = f(u)$ to get

$$\hat{f}(-n) = \frac{1}{2\pi} \int_{\pi}^{-\pi} f(-u) e^{-inu} d(-u) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(u) e^{-inu} du = \hat{f}(n).$$

Pick any $\varepsilon > 0$. Fejér's theorem tells us that there exists an $N \in \mathbb{N}$ such that

$$\sup_{\theta \in [-\pi, \pi]} |f(\theta) - F_N * f(\theta)| < \frac{\varepsilon}{2}.$$

Using $\hat{f}(n) = \hat{f}(-n)$, we get

$$\begin{aligned} D_k * f(\theta) &= \sum_{n=-k}^k \hat{f}(n) e^{in\theta} \\ &= \hat{f}(0) + \sum_{n=1}^k \hat{f}(n)(e^{in\theta} + e^{-in\theta}) \\ &= \hat{f}(0) + 2 \sum_{n=1}^k \hat{f}(n) \cos n\theta. \end{aligned}$$

But since $\cos n\theta = T_n(\cos \theta)$ for some polynomial T_n , we get that $D_k * f(\theta) = p_k(\cos \theta)$ for some polynomial p_k . So,

$$F_N * f(\theta) = \frac{1}{n} \sum_{k=0}^{n-1} D_k * f(\theta) = \frac{1}{n} (p_0 + \cdots + p_{n-1})(\cos \theta) = p(\cos \theta)$$

for some polynomial p . Now, for any $x \in [-1, 1]$ there exists $\theta \in [-\pi, \pi]$ such that $x = \cos \theta$. So

$$|g(x) - p(x)| = |g(\cos \theta) - p(\cos \theta)| = |f(\theta) - F_N * f(\theta)| < \frac{\varepsilon}{2}.$$

Finally we conclude that

$$\sup_{x \in [-1, 1]} |g(x) - p(x)| \leq \frac{\varepsilon}{2} < \varepsilon. \quad \square$$

4 Problem Set I

Solutions have been provided to some problems.

Problem 1. Let $\sum x_n$ be a series such that $x_n \geq 0$. Show that if $\sum x_n$ diverges, then it is not Abel summable.

Deduce that for a series with nonnegative terms, the different notions of convergence – usual, Cesàro, and Abel – are all equivalent.

Proof. Let $x_n \geq 0$ be such that it is Abel summable but not summable. Let s_n be its partial sums. Let $0 \leq r_k < 1$ be a sequence that is increasing to 1. Let

$$a_k = \sum_{n=0}^{\infty} x_n r_k^n.$$

Fix any $N \in \mathbb{N}$. We have

$$a_k \geq \sum_{n=0}^N x_n r_k^n.$$

Note that $\liminf a_k \geq 0$, since the terms a_k are nonnegative. Suppose $\liminf a_k < \infty$. Then

$$\liminf a_k \geq \liminf \sum_{n=0}^N x_n r_k^n = \lim_{k \rightarrow \infty} \sum_{n=0}^N x_n r_k^n = s_N.$$

But s_N can be arbitrarily large. This is a contradiction. Hence, $\liminf a_k$ is infinite, and x_n is not Abel summable.

If a series is not Abel summable then it is not Cesàro summable. Thus, a series of nonnegative terms either converges in all three notions, or diverges in all of them. $\square$

Problem 2. Conclude from problem 1 that if a series is divergent but Abel summable, then it must have infinitely many positive and infinitely many negative terms.

Proof. Suppose $\sum x_n$ is Abel summable but not summable. Let N be such that $x_n \geq 0$ for all $n \geq N$. Let $y_n = x_N, x_{N+1}, \dots$. Then y_n is not summable, since x_n is not summable. Therefore, by problem 1, y_n is not Abel summable. But this implies that x_n is not Abel summable. This is a contradiction. Hence, no such N can exist. That is, x_n has infinitely many negative terms.

Note that if $\sum x_n$ is Abel summable, then so is $\sum (-x_n)$. Likewise, if $\sum x_n$ is not summable, then neither is $\sum (-x_n)$. Repeat the argument on the sequence $-x_n$ to conclude that infinitely many terms of x_n are positive. $\square$

Problem 3. Observe that the conclusion of problem 2 holds if the condition of Abel summability is replaced by Cesàro summability.

Hint: If a series is Cesàro summable, then it is also Abel summable. So the conclusion holds.

Problem 4. Show that if a series is not Cesàro summable but is Abel summable, then it must have infinitely many positive and infinitely many negative terms.

Hint: If a series is not Cesàro summable, then it is not convergent. Given that the series is Abel summable, the conclusion holds.

Problem 5. Construct a series $\sum x_n$ such that its partial sums s_N are unbounded (that is $\limsup s_N = \infty$) but its Cesàro sum is 0.

Proof. Define the sequence s_n such that $s_0 = 0$ and

$$s_n = \begin{cases} k & \text{if } n = k^3 \\ 0 & \text{otherwise} \end{cases}.$$

Let x_n be the sequence whose partial sums are s_n . Clearly, s_n is unbounded. Note that the Cesàro partial sums σ_n of $\sum x_n$ are

$$\sigma_{n+1} = \frac{s_0 + \cdots + s_n}{n+1} = \frac{k(k+1)}{2(n+1)},$$

where $k = \left\lfloor n^{\frac{1}{3}} \right\rfloor$. Therefore,

$$0 \leq \sigma_{n+1} \leq \frac{n^{\frac{1}{3}}(n^{\frac{1}{3}} + 1)}{2n} \leq \frac{2n^{\frac{2}{3}}}{2n} = n^{-\frac{1}{3}}.$$

Hence σ_n converges to 0. □

Problem 6. Let $f: [-\pi, \pi] \rightarrow \mathbb{R}$ be continuous. Show that $\hat{f}(n) = \overline{\hat{f}(-n)}$. Conclude that the Fourier series of f can be alternatively written as the series

$$\sum_{n=0}^{\infty} (a_n \cos n\theta + b_n \sin n\theta).$$

Problem 7. Let L_n be the sequence of Lebesgue constants and let D_n be the Dirichlet kernel. Define a new family K_n where

$$K_n(t) = \frac{1}{L_n} |D_n(t)|.$$

Show that K_n is a Dirac family. Ponder what this means for the convergence of the Fourier series of $f \in C[-\pi, \pi]$.

Proof. One immediately sees that the maps K_n are positive, even and normalized. Now, for “peaking at 0”, fix any $\delta > 0$. Observe that

$$\int_{\delta < |t| < \pi} K_n(t) dt = \frac{1}{\pi L_n} \int_{\delta}^{\pi} \left| \frac{\sin(2n+1)\frac{t}{2}}{\sin \frac{t}{2}} \right| dt \leq \frac{1}{\pi L_n} \int_{\delta}^{\pi} \frac{1}{\sin \frac{t}{2}} dt \leq \frac{2(\pi - \delta)}{L_n \pi \sin \frac{\delta}{2}},$$

In the interval (δ, π) , $\sin \frac{t}{2}$ is positive and increasing. This is used in the last inequality. As n goes to infinity, so does L_n , and hence the integral goes to 0. □

Problem 8. Recall the centered Gaussian distribution

$$\mathcal{G}_\sigma(t) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(\frac{t}{\sigma}\right)^2\right].$$

Define the kernel G_n as

$$G_n(t) = \mathcal{G}_{\sqrt{1/2n}}(t).$$

Verify G_n is a Dirac family on $\mathbb{R}$.

Proof. The family G_n is positive, even and normalized. These are immediate. Chebyshev's concentration bound for a random variable X gives

$$\Pr[|X - \mathbb{E}X| > \delta] \leq \frac{\text{var } X}{\delta^2} \quad \text{for every } \delta > 0.$$

Let X be distributed according to G_n . In this case the above inequality says

$$\int_{|t|>\delta} G_n(t) dt \leq \frac{1}{2n\delta^2}.$$

Thus, as $n \rightarrow \infty$ the integral goes to 0. □

Problem 9. (Optional) From complex analysis, a simple consequence of Cauchy estimates is the following. Let $g: \mathbb{C} \rightarrow \mathbb{C}$ be an entire² function such that

$$|g(x + iy)| \leq \lambda_1 e^{\lambda_2 y^2}$$

for some positive numbers λ_1 and λ_2 . Let p_m be the Taylor series of g (around 0) truncated at the m th term. That is,

$$p_m(x) = \sum_{k=0}^m \frac{g^{(k)}(0)}{k!} x^k.$$

Then p_m satisfies

$$\sup_{x \in [-\pi, \pi]} |g(x) - p_m(x)| \leq C 2^{-m},$$

for some constant C . Use this and problem 8 to give another proof of Weierstrass approximation theorem. [This approach is close to Weierstrass's original proof].

2. An entire function $f: \mathbb{C} \rightarrow \mathbb{C}$ is a function that is differentiable at every $z \in \mathbb{C}$. This property is stronger than being a C^∞ function as a function of a real variable. However, for the scope of this question, those who are not familiar with complex analysis may assume “entire” to be the same thing as infinitely differentiable.

Proof. Let f be any continuous function on $[-\pi, \pi]$. Extend f to a continuous function on $\mathbb{R}$ with compact support. Let $g_n = f * G_n$. That is,

$$g_n(\theta) = \int_{-\pi}^{\pi} f(t) G_n(\theta - t) dt.$$

Then $g_n \rightarrow f$ uniformly. We assume that we can extend g_n to be defined on $\mathbb{C}$ (by replacing θ by any complex number z in the above equation), and that the Taylor series of g_n converges at every point (we verify that g_n at least has infinitely many derivatives). Then

$$|g_n(x + iy)| \leq \int_{-\pi}^{\pi} |f(t)| |G_n(\theta - t)| dt \leq \|f\| \int_{-\pi}^{\pi} \sqrt{\frac{n}{\pi}} |\exp[-n(x + iy - t)^2]| dt.$$

For any complex number z , we have $|e^z| = e^{\operatorname{Re} z}$. And

$$\operatorname{Re}(-n(x + iy - t)^2) = -n((x - t)^2 - y^2).$$

This gives

$$\begin{aligned} |g_n(x + iy)| &\leq \|f\| e^{ny^2} \int_{-\pi}^{\pi} \sqrt{\frac{n}{\pi}} \exp[-n(x - t)^2] dt \\ &= \|f\| e^{ny^2} \int_{-\pi}^{\pi} G_n(x - t) dt \\ &\leq \|f\| e^{ny^2}. \end{aligned}$$

Thus g_n satisfies the required conditions. Under these conditions we are told that the Taylor polynomial $p_m^{(n)}$ of g_n converges uniformly to g_n . Recall that g_n also converges uniformly to f . We begin by choosing g_N that is ε -close to f (in the supremum norm). For this g_N choose $p_m^{(N)}$ that is ε -close to g_N . We conclude the problem using

$$\|f - p_m^{(N)}\| \leq \|f - g_N\| + \|g_N - p_m^{(N)}\|. \quad \square$$

Problem 10. (Optional) We saw that Fejér's theorem can be used to derive Weierstrass approximation. Derive the same from Poisson's theorem.